Contents

Chapter 6 — Two Random Variables

6-1 Bivariate Distributions

6-1.1 Joint Distribution and Density

The joint (bivariate) distribution $F_{XY}(x, y)$ of two random variables X and Y is defined as the probability of the event $\{X \le x, Y \le y\}$:

$$F(x, y) = P\{X \le x, Y \le y\}$$

Properties of $F(x, y)$: 1. Boundary conditions: $F(-\infty, y) = 0$, $F(x, -\infty) = 0$, and $F(\infty, \infty) = 1$. 2. Strip probabilities: The probability in a vertical or horizontal half-strip is evaluated as:

$$P\{x_1 < X \le x_2, Y \le y\} = F(x_2, y) - F(x_1, y)$$

$$P\{X \le x, y_1 < Y \le y_2\} = F(x, y_2) - F(x, y_1)$$

3. Rectangle probability (Inclusion-Exclusion):

$$P\{x_1 < X \le x_2, y_1 < Y \le y_2\} = F(x_2, y_2) - F(x_1, y_2) - F(x_2, y_1) + F(x_1, y_1) \ge 0$$

Joint Density: The joint density function is defined via partial differentiation:

$$f(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y}$$

The probability mass in any region D is the integral of the joint density over that region:

$$P\{(X, Y) \in D\} = \iint_D f(x, y)\, dx\, dy$$

6-1.2 Marginal Statistics

The marginal distributions and densities are obtained by integrating out the other variable:

$$F_X(x) = F(x, \infty), \quad F_Y(y) = F(\infty, y)$$

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$$

Example 6-1 (Probability Mass in a Circle) Suppose $f(x,y) = \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/2\sigma^2}$. Find the probability mass m in the circle $x^2 + y^2 \le a^2$. **Solution:** Convert to polar coordinates ($x = r\cos\theta, y = r\sin\theta$).

$$m = \frac{1}{2\pi\sigma^2} \int_0^a \int_{-\pi}^{\pi} e^{-r^2/2\sigma^2} r\,dr\,d\theta = 1 - e^{-a^2/2\sigma^2}$$

6-1.3 Independence

Random variables X and Y are independent if $P\{X \in A, Y \in B\} = P\{X \in A\}P\{Y \in B\}$. Equivalently:

$$F(x,y) = F_X(x)F_Y(y) \quad \text{and} \quad f(x,y) = f_X(x)f_Y(y)$$

Example 6-2: Buffon's Needle A fine needle of length $2a$ is dropped on a board with parallel lines distance $2b$ apart ($b > a$). Determine the probability p that the needle intersects a line.

Solution: Let X be the distance from the center to the nearest line, $X \sim U(0,b)$. Let Θ be the angle with the perpendicular, $\Theta \sim U(0, \pi/2)$. Assuming independence: $f(x,\theta) = \frac{1}{b}\frac{2}{\pi} = \frac{2}{\pi b}$. Intersection occurs if $X < a\cos\Theta$. Integrating the joint density over this region:

$$p = P\{X < a\cos\Theta\} = \frac{2}{\pi b}\int_0^{\pi/2} a\cos\theta\,d\theta = \frac{2a}{\pi b}$$

Theorems on Independence: * **Theorem 6-1:** If X and Y are independent, then the random variables $Z = g(X)$ and $W = h(Y)$ are also independent. * **Theorem 6-2:** If experiments S_1 and S_2 are independent, random variables defined on them are independent.

6-1.4 Joint Normality & Circular Symmetry

Two variables are **jointly normal** if their density is the exponential of a negative quadratic:

$$f(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\eta_X)^2}{\sigma_X^2} - 2\rho\frac{(x-\eta_X)(y-\eta_Y)}{\sigma_X\sigma_Y} + \frac{(y-\eta_Y)^2}{\sigma_Y^2}\right]\right\}$$

Note: If two variables are jointly normal, their marginals are normal. However, the converse is not true.

Example 6-3: Marginally Normal but NOT Jointly Normal Consider the function:

$$f(x,y) = f_X(x)f_Y(y)[1 + \rho\{2F_X(x) - 1\}\{2F_Y(y) - 1\}] \quad (|\rho| < 1)$$

If $f_X(x)$ and $f_Y(y)$ are standard normal densities, integrating this $f(x,y)$ over y yields exactly $f_X(x)$ (since the integral of the odd term over the entire real line vanishes). Thus, the marginals are perfectly normal, but the joint PDF clearly isn't jointly normal.

- **Theorem 6-3 (Circular Symmetry):** If X and Y are independent and have a circularly symmetrical joint density ($f(x,y) = g(\sqrt{x^2+y^2})$), then they *must* be normal with zero mean and equal variance.

6-1.5 Discrete Type and Line Masses

For discrete variables, $P\{X = x_i, Y = y_k\} = p_{ik}$. Marginals are $p_i = \sum_k p_{ik}$ and $q_k = \sum_i p_{ik}$.

> **Example 6-4 & 6-5: Dice Tosses** * **(a)** Toss a fair die. X is the dots shown, $Y = 2X$. The masses exist only on 6 points $(i, 2i)$, each with probability 1/6. * **(b/c)** Toss a die twice. If $X = |i - k|$ and $Y = i + k$, the masses are distributed over 21 valid coordinate points out of the total $6 \times 11 = 66$ plane points. * **(Example 6-5)** If X is the first toss and Y is the second, they are independent, so $p_{ik} = p_i p_k$.

6-4 Joint Moments

6-4.1 Definitions and Linearity

Theorem 6-4: The expected value of a function $g(X, Y)$ is evaluated directly using the joint density:

$$E\{g(X,Y)\} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f(x,y)\, dx\, dy$$

Due to linearity, $E\{X + Y\} = E\{X\} + E\{Y\}$, but generally $E\{XY\} \neq E\{X\}E\{Y\}$.

6-4.2 Covariance, Correlation, and Orthogonality

- **Covariance:** $C_{XY} = E\{(X - \eta_X)(Y - \eta_Y)\} = E\{XY\} - E\{X\}E\{Y\}$.
- **Correlation Coefficient:** $\rho_{XY} = \frac{C_{XY}}{\sigma_X \sigma_Y}$, bounded by $|\rho_{XY}| \leq 1$.
- **Uncorrelated:** If $C_{XY} = 0$ (implies $E\{XY\} = E\{X\}E\{Y\}$).
- **Orthogonal:** If $E\{XY\} = 0$ (denoted as $X \perp Y$).
- **Cosine Inequality:** Treating variables as vectors, $(E\{XY\})^2 \leq E\{X^2\}E\{Y^2\}$.

> **Example 6-30: Correlation of Jointly Normal RVs** For jointly normal variables, evaluating the integral to find the covariance yields exactly $C_{XY} = \rho\sigma_X\sigma_Y$. Therefore, the parameter ρ in the joint normal formula is exactly the correlation coefficient.

6-4.3 Theorem 6-5: Independence vs. Uncorrelatedness

If two random variables are independent, they are uncorrelated ($E\{XY\} = E\{X\}E\{Y\}$). **The converse is generally false**, except for jointly normal random variables.

> **Example 6-31: Uncorrelated but NOT Independent** Let $X \sim U(0,1)$ and $Y \sim U(0,1)$ be independent. Define $Z = X + Y$ and $W = X - Y$. 1. Check Correlation: $E\{ZW\} = E\{(X+Y)(X-Y)\} = E\{X^2 - Y^2\} = E\{X^2\} - E\{Y^2\} = 0$. Since $E\{Z\} = 0.5 + 0.5 = 1$ and $E\{W\} = 0.5 - 0.5 = 0$, the covariance $C_{ZW} = E\{ZW\} - E\{Z\}E\{W\} = 0 - (1)(0) = 0$. Thus, Z and W are **uncorrelated**. 2. Check Independence: Using transformation limits, $f_{ZW}(z,w) = 1/2$ inside a diamond-shaped region and 0 elsewhere. The marginals are $f_Z(z)$ (a triangle) and $f_W(w) = 1 - |w|$. Since $f_{ZW}(z,w) \neq f_Z(z)f_W(w)$, they are **not independent**.

6-4.4 Variance of a Sum and Moments

$$\sigma_{X+Y}^2 = \sigma_X^2 + \sigma_Y^2 + 2\rho_{XY}\sigma_X\sigma_Y$$

If uncorrelated, the variance of the sum is the sum of the variances.

> **Example 6-32: Linear Combination of Normal RVs** If X and Y are jointly normal with $\eta_X = 10, \eta_Y = 0, \sigma_X^2 = 4, \sigma_Y^2 = 1, \rho_{XY} = 0.5$. Find the joint density of $Z = X+Y$ and $W = X-Y$. **Solution:** Linear combinations of normal RVs are jointly normal. Means: $\eta_Z = 10 + 0 = 10$, $\eta_W = 10 - 0 = 10$. Variances: $\sigma_Z^2 = 4 + 1 + 2(0.5)(2)(1) = 7$. $\sigma_W^2 = 4 + 1 - 2(0.5)(2)(1) = 3$. To find correlation, use $E\{ZW\} = E\{X^2 - Y^2\}$. We know $E\{X^2\} = \sigma_X^2 + \eta_X^2 = 4 + 100 = 104$ and

$E\{Y^2\} = 1 + 0 = 1$. Thus, $E\{ZW\} = 104 - 1 = 103$. Covariance $C_{ZW} = E\{ZW\} - \eta_Z \eta_W = 103 - 100 = 3$. $\rho_{ZW} = \frac{3}{\sqrt{7\times 3}} = \sqrt{3/7}$. Thus, $Z, W \sim N(10, 10, 7, 3, \sqrt{3/7})$.

6-6 Conditional Distributions

6-6.1 Conditional Density Definition

For a point condition $X = x$, the conditional density is defined via a limit and yields:

$$f(y|x) = \frac{f(x,y)}{f_X(x)} \quad \text{and} \quad f(x|y) = \frac{f(x,y)}{f_Y(y)}$$

Example 6-40: Triangle Distribution Given $f(x,y) = k$ for $0 < x < y < 1$. Find $f(x|y)$ and $f(y|x)$. **Solution:** First, find k: $\int_0^1 \int_0^y k\,dx\,dy = \int_0^1 ky\,dy = k/2 = 1 \implies k = 2$. Marginals: $f_X(x) = \int_x^1 2\,dy = 2(1-x)$. $f_Y(y) = \int_0^y 2\,dx = 2y$. Conditionals:

$$f(x|y) = \frac{2}{2y} = \frac{1}{y}, \quad 0 < x < y < 1$$

$$f(y|x) = \frac{2}{2(1-x)} = \frac{1}{1-x}, \quad 0 < x < y < 1$$

Example 6-41: Conditional of Joint Normal If X and Y are jointly normal, factoring $f_X(x)$ out of the joint PDF reveals that $f(y|x)$ is also a normal density. Specifically, it has:

$$\text{Mean: } \eta_{y|x} = \eta_Y + \rho\sigma_Y \frac{x - \eta_X}{\sigma_X} \quad \text{Variance: } \sigma^2_{y|x} = \sigma^2_Y(1-\rho^2)$$

6-6.2 Bayes' Theorem for Densities

Using the total probability expansion $f_Y(y) = \int f(y|x) f_X(x) dx$, we obtain Bayes' theorem:

$$f(x|y) = \frac{f(y|x) f_X(x)}{\int_{-\infty}^{\infty} f(y|x) f_X(x)\,dx}$$

Example 6-42: Phase in Noise (Bayes Application) An unknown phase $\Theta \sim U(0, 2\pi)$ is transmitted. We observe $R = \Theta + N$, where $N \sim N(0, \sigma^2)$ is independent noise. Determine $f(\theta|r)$. **Solution:** 1. Prior: $f_\Theta(\theta) = 1/(2\pi)$. 2. Likelihood given θ: Since $R = \Theta + N$, $f(r|\theta) \sim N(\theta, \sigma^2)$. 3. Posterior via Bayes:

$$f(\theta|r) = \frac{e^{-(r-\theta)^2/2\sigma^2}}{\int_0^{2\pi} e^{-(r-\theta)^2/2\sigma^2} d\theta} \quad \text{for } 0 < \theta < 2\pi$$

Conclusion: The flat prior $U(0, 2\pi)$ becomes a bell-shaped curve peaking exactly at the observation r.

6-6.3 Discrete Conditoning (Markov Matrices & Sums)

For discrete variables, the conditional probability $\pi_{ik} = P\{Y = y_k | X = x_i\} = \frac{p_{ik}}{p_i}$ forms a **Markov Matrix** (nonnegative elements, rows sum to 1).

Example 6-43: Sum of Binomials/Poissons * **Binomials:** If $X \sim \text{Bin}(m, p)$ and $Y \sim \text{Bin}(n, p)$ are independent, then $X + Y \sim \text{Bin}(m + n, p)$. The conditional distribution $P\{X = x | X + Y = x + y\}$ is strictly **hypergeometric**: $\frac{\binom{m}{x}\binom{n}{y}}{\binom{m+n}{x+y}}$. * **Poissons:** If $X \sim P(\lambda)$ and $Y \sim P(\mu)$ are independent, the sum is $P(\lambda + \mu)$. The conditional distribution $P\{X = k | X + Y = n\}$ is strictly **binomial**: $\binom{n}{k}(\frac{\lambda}{\lambda+\mu})^k(\frac{\mu}{\lambda+\mu})^{n-k}$.

6-6.4 System Reliability and Failure Rates

Let X be the time to failure. Reliability $R(t) = 1 - F(t)$. The **conditional failure rate (hazard rate)** $\beta(t)$ is the probability density of failure exactly at t, given survival up to t:

$$\beta(t) = f(t|X > t) = \frac{f(t)}{1 - F(t)}$$

This allows us to write $f(x) = \beta(x)\exp\{-\int_0^x \beta(t)dt\}$.

> **Example 6-44, 6-45, 6-46: Memoryless Systems** A system is **memoryless** if it is "as good as new" regardless of how long it ran: $f(x|X > t) = f(x - t)$. Using $\beta(t)$, this implies $\beta(t)$ is a constant c. Integrating yields $f(x) = ce^{-cx}$. Thus, a system is memoryless **if and only if** its failure time is exponentially distributed.

> **Example 6-47: Weibull Density** A common hazard rate is $\beta(t) = ct^{b-1}$. Using the integral formula, this generates the Weibull density: $f(x) = cx^{b-1}\exp\{-cx^b/b\}$.

Interconnection of Systems: * **Parallel** (Fails if BOTH fail): $Z = \max(X, Y)$. * **Series** (Fails if EITHER fail): $W = \min(X, Y)$. * **Standby** (Second starts when first fails): $S = X + Y$.

6-7 Conditional Expected Values

6-7.1 Conditional Mean and Regression

$$E\{g(Y)|M\} = \int_{-\infty}^{\infty} g(y)f(y|M)\,dy$$

The function $\varphi(x) = E\{Y|X = x\}$ is defined as the **regression line**.

> **Example 6-48: Finding Conditional Expected Values** Given $f(x, y) = 1$ for $0 < |y| < x < 1$. Find $E\{X|Y = y\}$ and $E\{Y|X = x\}$. **Solution:** Find marginals: $f_X(x) = \int_{-x}^{x} 1\,dy = 2x$, $f_Y(y) = \int_{|y|}^{1} 1\,dx = 1 - |y|$. Form conditionals: $f(x|y) = 1/(1 - |y|)$ and $f(y|x) = 1/2x$. Integrate to find expected values:
>
> $$E\{X|y\} = \int_{|y|}^{1} \frac{x}{1 - |y|}\,dx = \frac{1 + |y|}{2}$$
>
> $$E\{Y|x\} = \int_{-x}^{x} \frac{y}{2x}\,dy = 0$$

> **Example 6-49: Galton's Law (Regression toward the mean)** If X (parent height) and Y (child height) are identically distributed and positively correlated ($\rho > 0$), Galton noted that $E\{Y|X = x\}$ lies closer to the mean η than the parent's value x.
>
> For jointly normal RVs, $E\{Y|x\} = \eta + \rho(x - \eta)$. Since $\rho < 1$, the slope pulls extreme values of X back toward the population mean η.

6-7.2 Nested Expectations (Law of Total Expectation)

Since $E\{Y|X\}$ is a function of the random variable X, it is itself a random variable. The mean of this conditional expectation equals the unconditional mean:

$$E\{E\{Y|X\}\} = E\{Y\}$$

More generally, knowns given X can be factored out:

$$E\{g_1(X)g_2(Y)|X\} = g_1(X)E\{g_2(Y)|X\}$$

Example 6-50: Finding Joint Normal Moments using Nested Expectations Let $X, Y \sim N(0, 0, \sigma_1^2, \sigma_2^2, \rho)$. Prove that $E\{X^2Y^2\} = E\{X^2\}E\{Y^2\} + 2(E\{XY\})^2$. **Solution:** 1. Use nested expectations: $E\{XY\} = E\{XE\{Y|X\}\}$. Since $E\{Y|X\} = \rho\sigma_2 X/\sigma_1$, we get $E\{XY\} = \rho(\sigma_2/\sigma_1)E\{X^2\} = \rho\sigma_1\sigma_2$. 2. For the 4th-order moment: $E\{X^2Y^2\} = E\{X^2E\{Y^2|X\}\}$. 3. Recall $E\{Z^2\} = \text{Var}(Z) + (E\{Z\})^2$. Thus: $E\{Y^2|X\} = \sigma_{y|x}^2 + (E\{Y|X\})^2 = \sigma_2^2(1-\rho^2) + \left(\frac{\rho\sigma_2 X}{\sigma_1}\right)^2$. 4. Substitute this back into the outer expectation:

$$E\{X^2Y^2\} = E\left\{X^2\left[\sigma_2^2(1-\rho^2) + \rho^2\frac{\sigma_2^2}{\sigma_1^2}X^2\right]\right\}$$

5. Distribute X^2. Using the normal property $E\{X^4\} = 3\sigma_1^4$, this simplifies to:

$$E\{X^2Y^2\} = \sigma_2^2(1-\rho^2)\sigma_1^2 + \rho^2\frac{\sigma_2^2}{\sigma_1^2}(3\sigma_1^4) = \sigma_1^2\sigma_2^2 + 2\rho^2\sigma_1^2\sigma_2^2$$

Which is exactly $E\{X^2\}E\{Y^2\} + 2(E\{XY\})^2$.